

DAY 4

Pattern Problems & Nested Loops



DSA Preparation using
Python & Java



Master patterns and nested loops
to solve real-world DSA problems.



Nested
Loops



Rows &
Columns



Pattern
Printing



$O(n^2)$



Python
+ Java

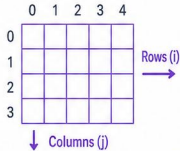
PATTERN PRINTING

```
  *
 * *
* * *
* * * *
* * * * *
```

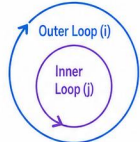
```
* * * * *
* * * * *
  * * *
    * *
      *
```

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

ROWS & COLUMNS (GRID)



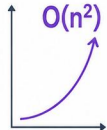
NESTED LOOPS



NESTED LOOPS (EXAMPLE)

```
1 for i in range(n):
2   for j in range(n):
3     print("*", end=" ")
4   print()
5 Time Complexity:  $O(n^2)$ 
```

```
* * * * *
* * * * *
* * * * *
* * * * *
```



Day-4

07/05/2026.

→ The main goal of Day 4 is to understand nested loops and use them to solve pattern problems.

① What is a Nested loop?

A nested loop makes (or) means a loop inside another loop

Eg:

For each row:

For each Column:

print something.

* * *
* * *
* * *

→ In pattern problems:

↳ Outer loops → Control rows

↳ Inner loops → Control columns

Example pattern

This has:

* * *
* * *
* * *

3 rows

3 columns

→ So we use one loop for rows & one loop for columns

3x3

② Why Nested Loops are Important?

→ Nested loops are important because they help us solve problems where we need to compare (or) process pairs of values

Eg: → Printing Pattern.

→ Matrix problems

→ Comparing every element with every other element.

→ 2D arrays

→ Brute force problems.

→ Nested loops are also common in DSA, but we must be careful because they usually increase time complexity.

③ Outer Loop and Inner Loop.

Eg. For this pattern:

```
* * *
* * *
* * *
```

We can think like this:

Row 1 → print 3 stars.

Row 2 → print 3 stars.

Row 3 → print 3 stars.

So:

Outer Loop → runs 3 times for rows.

Inner Loop → runs 3 times for stars in each row.

Total operations

$$3 \times 3 = 9$$

Python

```
for i in range(3):
    for j in range(3):
        print("*", end=" ")
    print()
```

→ i controls rows
→ j controls columns
end=" ", prints stars on
same line.
print() moves to next
line.

Java Example

```
class Main{
```

```
    public static void main (String[] args) {
```

```
        for (int i = 1; i <= 3; i++) {
```

```
            for (int j = 1; j <= 3; j++) {
```

```
                System.out.print("x ");
```

```
            }
```

```
        System.out.println();
```

```
    }  
}
```

→ Time Complexity $\rightarrow O(n \times n) \rightarrow O(n^2)$

→ Space complexity $\rightarrow O(1)$

→ ~~Beac~~ Because we not using extra memory that grows with input